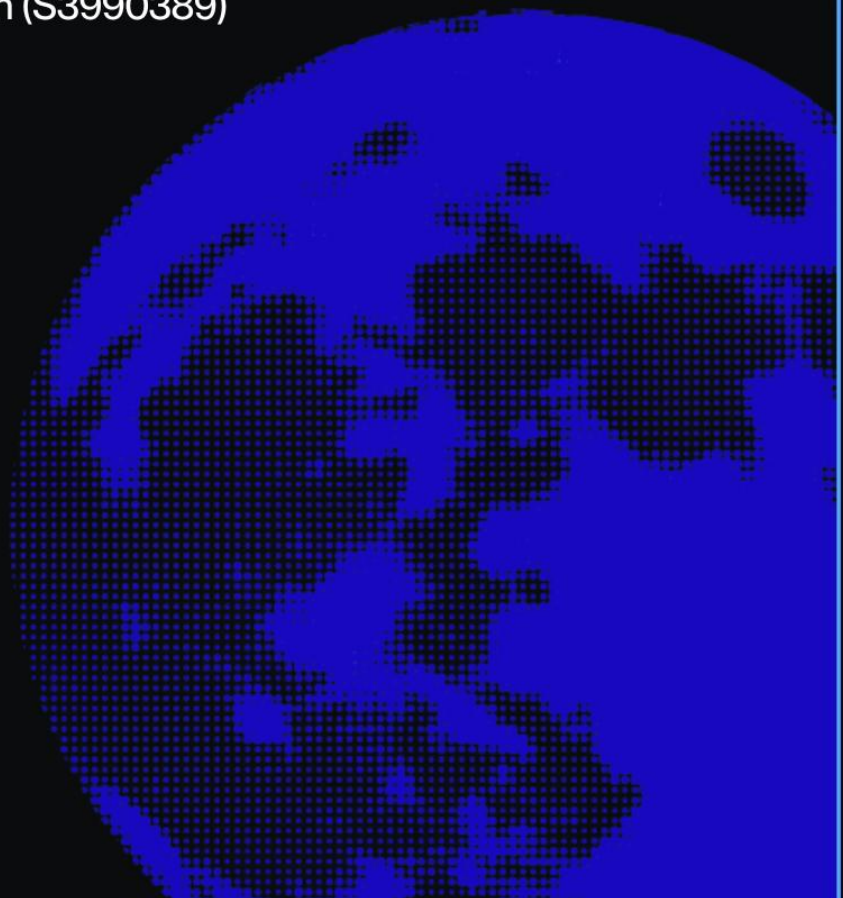


Assignment 3
Business Proposal

A BLOCKCHAIN-BASED *SUPPLY CHAIN FINANCE* PROTOCOL

Lecturer: Dr. Binh Nguyen

Student: Ngo Phuc Thinh (S3990389)





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Student Name	Ngo Phuc Thinh
Student ID	S3990389
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I declare that in submitting all work for this assessment, I have read, understood, and agree to the content and expectations of the Assessment Declaration.

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I. INTRODUCTION

Vietnamese banks have built substantial SCF infrastructure, especially HDBank, which alone runs a VND 40 trillion (USD 1.6 billion) SCF portfolio across 4,202 tier-1 suppliers (VnEconomy 2025). However, Vietnam's factoring-to-trade ratio stands at 0.15%, compared to 8% in Mexico, and only 9.3% of SMEs reach formal bank credit (FiinGroup 2025; Deng 2025). Supply chain volume flows through tier-2 and tier-3 firms, but banks cannot verify their invoices, identities, and order trails because they aren't crossing institutional boundaries, and therefore lend to them. A permissioned blockchain SCF protocol operating under the VIFC sandbox is proposed, designed to make tier-2/3 receivables verifiable across banks without placing any centralized party.

II. EMPATHIZE

1. TARGET CUSTOMERS

The target customers are tier-2/3 Vietnamese suppliers whose receivables banks cannot verify. They are usually owner-operator firms with 10-50 employees and annual revenue of VND 7.5-40 billion, located in industrial corridors such as Bac Ninh and Dong Nai, across electronics, textile, and seafood manufacturing chains (FiinGroup 2025).

The composite persona used throughout this report is Mr. Thang Ngo, grounded in cases like Hanpo Vina JSC (Hoang 2021) (Appendix B1.1). The running deal example will be Mr. Thang's VND 200 million invoice to an anchor (Appendix A1).

Reaching this supplier requires SCF-active commercial banks to advance the cash. HDBank's VCF platform marks the ceiling of single-bank reach, financing 4,202 tier-1 suppliers, but stopping at tier-1 because going deeper requires cross-bank verification (VnEconomy 2025). Moreover, HDBank faces external pressure, as its USD 215 million syndicated loan from FinDev Canada, SMBC, and JICA ties disbursement to data transparency conditions (VNBA 2025).

2. PAIN IDENTIFIED

Several key pain points are identified in the tier-2/3 supplier journey: credit invisibility, the informal-loan trap, invoice fraud and double-pledge risk, slow PO-to-cash conversion, and verification gaps (Appendix B2; B6).

From anchor buyers to tier-3 suppliers, it can take 90-120+ days for capital to cascade, creating slow PO-to-cash conversion (Le 2026). Due to these timing issues, tier-2/3 suppliers face a cash gap at delivery and fall into an informal-loan trap at an annualized rate of around 22%, eroding 40% net profit (Archer et al. 2020).

The formal alternative remains inaccessible due to credit invisibility and fraud risk. Regarding credit invisibility, it has affected 40% of five-year-old SMEs that have never received a bank loan, as credit history cannot be built without a banking record (FiinGroup 2025). Second, when banks want to lend, they cannot detect whether the same invoice is already pledged elsewhere. In the First Brands case, they collapsed in 2025 with USD 2.3 billion in factoring liabilities under double-financing investigation (Global Trade Review 2025). These invoice fraud and double-pledge risks prompted VNBA to call for blockchain in invoice management (VNBA 2026a).

These lead to a verification gap as the root. No hard collateral and no shared registry enable aggregate verification, leaving suppliers stuck in the loop where no collateral, no credit, no growth (Le and Truong 2026; Global Trade Review 2025).

III. DEFINE

1. PROBLEM STATEMENT

Tier-2/3 suppliers in Vietnam cannot convert their purchase orders into working capital through formal banking; therefore, they rely on informal loans, which eat up 40% of net profit (Archer et al. 2020). Despite existing regulations, including Resolution 68 and Circular 20/2024, banks still cannot reach these suppliers because their invoice authenticity and credit history are limited by institutional boundaries. All of this results in a 9.3% formal-credit access rate for SMEs (FiinGroup 2025). The risk of inaction is severe, as EU CBAM and CSDDD push traceability requirements down to deep-tier suppliers, and unverifiable Vietnamese exporters risk being dropped from European supply chains (UB Speeda 2026).

2. ROOTCAUSE ANALYSIS

Five-Whys-fishbone analysis (Appendix B3) shows four reinforcing drivers explain why the verification gap persists:

Banks underwrite tier-1 because anchor identities are visible; at tier-2/3, visibility collapses, and lenders default to asymmetric-information rules that price for the worst case or step away from the segment (Dong et al. 2023; Liu 2026).

Incentives are also misaligned, as Wang et al. (2023) find that anchor and tier-1 firms gain commercially from withholding data, including capturing more trade credit from suppliers, protecting IP, and increasing negotiating power (Kembro 2025).

Regarding regulation, a National Assembly member confirms SMEs still face credit barriers a year after Resolution 68 because of unspecified operating procedures, registry standards, and dispute mechanisms (Vietnam.vn 2026).

Finally, any operator owning the platform gains a pricing view rivals cannot match, raising governance concerns and deterring participation, which also caused TradeLens's collapse (unclear governance and confidentiality concerns) (Najati 2025; Ruff 2024).

3. HOW MIGHT WE

From these four drivers, four main How Might We directions are proposed:

- HMW make tier-3 receivable verifiable across banks, with the credit history that follows from it portable?
- HMW make participation a commercial gain for anchors and tier-1 firms, so disclosing downstream data lowers their own supply risk?
- HMW run the verification layer as shared infrastructure that no single bank owns or extracts pricing intelligence from?
- HMW turn Resolution 68 and Circular 20/2024 into a bankable operating procedure with a defined registry standard and dispute mechanism?

IV. IDEATE

3. IDEA DESCRIPTION

The proposed solution is a permissioned protocol-style infrastructure operating under VIFC sandbox status, not a single-owner platform. The five-component technical stack (Besu/QBFT, ERC-3643, ONCHAINID, Chainlink Functions, CascadeEngine) uses MISA meInvoice as a regulated e-invoice

source and aims to reduce tier-2/3 supplier financing costs from 22% informal to 8-9% formal (MISA 2025).

HMW Direction	Protocol Design
Verifiable receivable and portable credit across banks	Consortium-governed permissioned chain, ERC-3643 token enforcing one claim per receivable, and ONCHAINID carrying credit signal across banks
Commercial gain for anchors and tier-1 firms, so disclosure lowers their own supply risk	CascadeEngine atomic settlement compresses the anchor's working-capital cycle in exchange for tier-2/3 visibility
Run the verification layer as shared infrastructure that no single bank owns or extracts pricing intelligence from	QBFT consensus among bank nodes under VIFC sandbox neutrality, the bank verifies its state independently
Resolution 68 and Circular 20/2024 turned into a bankable operating procedure	ERC-3643 token uniqueness provides the registry standard, ONCHAINID provides the identity layer, and consortium voting plus VIFC arbitration provide the dispute mechanism

Table 1: HMW mapped with Protocol Design

4. THEORATICAL SUPPORT

Institutional economics justifies the consortium-governance choice. Long-lived common-pool institutions survive because users affected by operating rules also shape them, which a profit-conflicted operator cannot honor (Ostrom 1990). VIFC and VNBA sponsorship put rule-making with the banks that bear the credit risk.

Transaction cost economics and game theory explain the bank's commercial calculus. Cross-tier financing creates a three-way equilibrium in which banks reach unbankable tier-2/3 segments, anchors reduce supply risk, and suppliers gain access to compressed-rate credit (Dong et al. 2023). Building it

in-house requires bilateral agreements scaling quadratically, the make-vs-buy barrier (Williamson 1985).

V. PROTOTYPE

1. KEY FEATURES

The prototype closes the verification gap by chaining five components across one deal lifecycle, each addressing a specific pain theme (Section 2.2) through documented mechanisms rather than generic capabilities.

Component	Pain Theme Addressed	Mechanism
Chainlink Functions plus Automation	Invoice authenticity	Pulls invoice data from MISA meInvoice (Decree 70/2025 e-invoice source) before any token is minted, so the chain verifies against the source rather than the supplier's word (Chainlink 2026).
ERC-3643 permissioned token	Double-pledging risks	Each receivable becomes a digital certificate that only one holder can carry, with transfers gated by identity and compliance checks (Lebrun et al. 2021)
ONCHAINID identity layer	Credit invisibility	A reusable digital ID carrying verified KYC and repayment history, so a second bank inherits the record (Tokeny 2024)
Hyperledger Besu with QBFT consensus	Verification gap	A private blockchain shared among bank validators, where every record requires a two-thirds supermajority vote so that no single party can fabricate one (Hyperledger Foundation 2026)
CascadeEngine smart contract	Slow PO-to-cash covenversion	When the anchor pays at maturity, it routes settlement to the bank that financed earlier (Liao 2021), and the supplier's repayment record cascades as credit signals (Shu et al. 2024)

Table 2: Problem-Solution fit

2. ARCHITECTURAL DESGIN

The validator banks own the consensus base, VIFC and VNBA own rule-setting, the operator owns only the middle layers, wiring those two together, and regulated off-chain sources (MISA, Open Banking) stay outside the chain. No single party controls more than one stratum, which makes the consortium auditable rather than custodial.

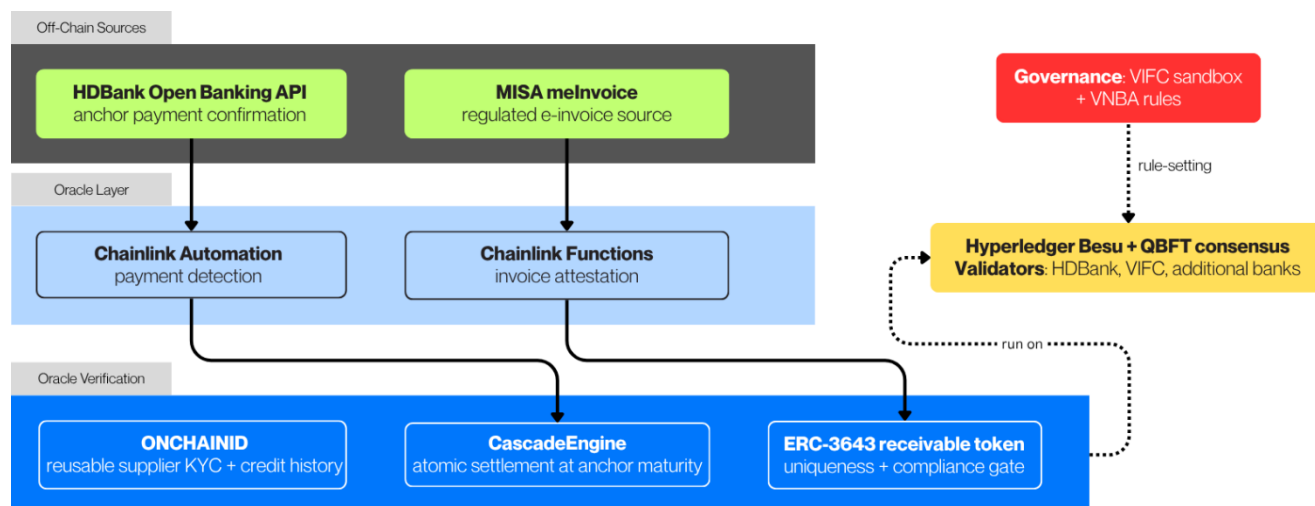


Figure 1: Prototype architecture (Source: Author)

3. BUSINESS MODEL

The operator is a **B2B SaaS** startup maintaining the protocol's smart contract and oracle integrations within the VIFC sandbox. Bank nodes maintain consensus; VIFC and VNBA set control rules; and the operator holds no transaction data or credit risk.

Revenue will mainly come from banks' annual subscription (USD 15K in Year 1, rising to USD 30-40K from Year 3 as the consortium grows). Also, a 12-15 basis-point take rate is charged on CascadeEngine settlements, following Solomon's (2015) "rules and rails" tier of 0-20 bps, one step below SCF financing-tier peers like C2FO at 50-250 bps, because what the protocol sells is verification, not capital (Appendix A2). Tier-2/3 suppliers, anchors, and regulators are not charged. They are compensated instead by rate compression, supply-risk reduction, and sandbox endorsement, respectively.

This onboarding-low pricing rising with activity responds to the Marco Polo case, where 30 banks signed up but never transacted because fees were front-loaded during onboarding (Wragg 2023a). The take rate fires only after deals clear, so earnings rise with consortium thickness rather than sign-up count.

According to Appendix A1.3, the per-deal value is splitted 81% to the supplier, 16% to the financing bank, and 3% to the operator, keeping the supplier the dominant beneficiary while the bank keeps margin above its cost of funds.

Comparable models occupy different structural positions within the SCF ecosystem. For example, TReDS India is a multi-bank invoice marketplace without cross-tier cascade settlement, whereas NAFIN Mexico required state-development-bank ownership, a sandbox startup cannot replicate (M1xchange 2023; Klapper 2006). Neutrality is what makes this configuration effective, since only a non-bank operator can manage a verification layer that rival banks accept, rather than fragment it.

4. USER JOURNEY

Mr. Thang Ngo's running deal will be used for visualizing, where he issues a VND 200M invoice via MISA meInvoice, Chainlink attests issuance, an ERC-3643 token mints to his ONCHAINID, HDBank's validator advances 95% same-day, and CascadeEngine settles atomically as the anchor pays 90 days later.

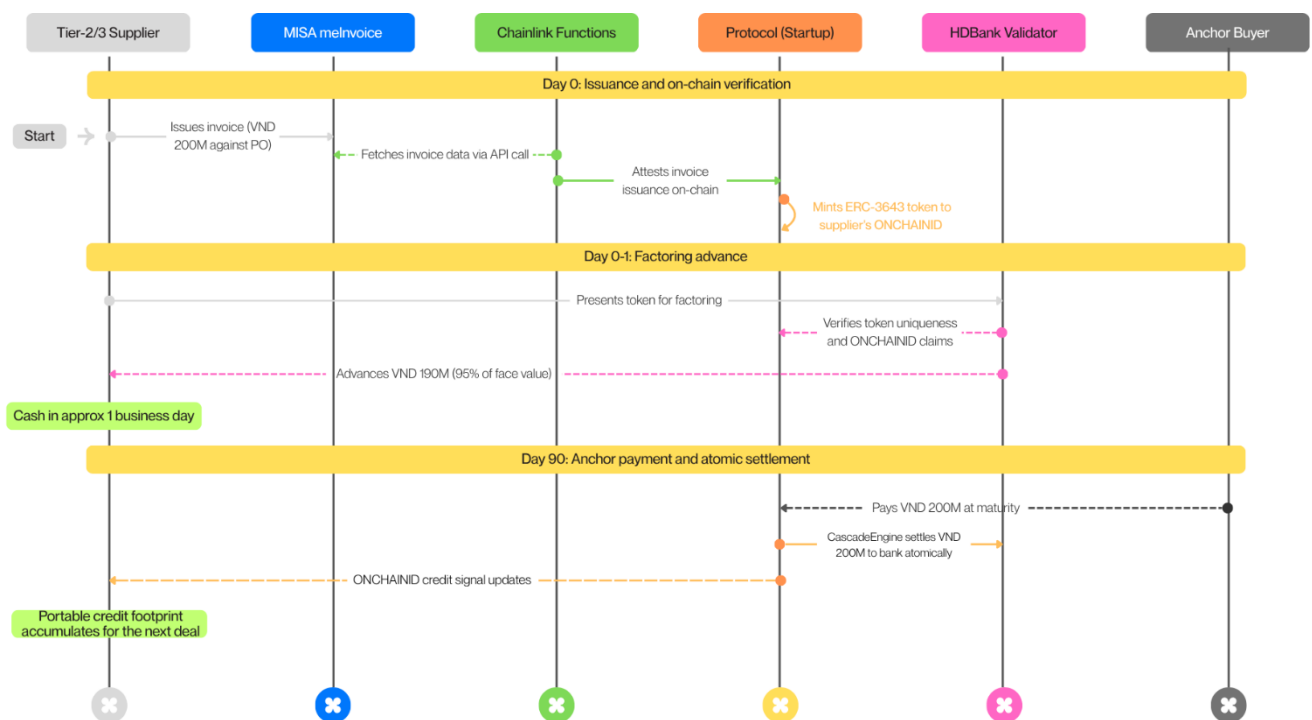


Figure 2: Prototype user interaction flow (Source: Author)

5. TEAM FORMATION AND COMMUNITY ENGAGEMENT

Four roles anchor the founding team: tech lead (smart-contract architecture), regulatory lead (sandbox compliance under Decree 94/2025), banking-relationships lead (validator onboarding), and SCF domain expert (product credibility) (Hoang and Keil 2025).

The validator-bank consortium is the growth unit. Each new bank brings its anchor portfolio and tier-2/3 chains, so deal flow scales with consortium membership, not sales. The protocol cannot be operator-owned because banks will not place sensitive transaction data on infrastructure controlled by a rival or startup, the constraint that killed prior consortium platforms (Jovanovic et al. 2022). VNBA TFR's eight-bank working group shows the pattern is viable in the Vietnamese sector (VNBA 2026b).

VI. TESTING

1. MVP AND MVE

The MVP runs under the VIFC sandbox to test whether HDBank, as the financing bank, can underwrite around 20 tier-2/3 suppliers once the verification gap is closed.

The MVE comprises six actors who validate the protocol's commercial assumption, each with a distinct market role and gain (Lewrick 2018).

Actor	Market role	Gain
Operator startup	Smart-contract and oracle maintainer	Subscription (USD 15K Year 1) plus 12-15bps take rate on settled volume (appendix A2)
HDBank	First financing bank	Access into deeper tier supplier; satisfies 2025 syndicated loan data-transparency conditions (VNBA 2025)
Anchor Buyer	First buyer counterparty	Tier-2/3 liquidity lowers delivery-default exposure on the anchor's production line
20 tier-2/3 suppliers	First credit consumers	Financing rate reduced from 22% to 8-9% formal, plus an ONCHAINID credit history (appendix A1)
VIFC	Sandbox host	Market formalization under the sandbox's regulatory mandate (Decree 323/2025)

VNBA	Consortium pattern endorser	Operating template for the Trade Finance Registry consortium piloted across eight banks (VNBA 2026b)
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Table 3: Actors' gains mapping

2. GO-TO-MARKET

HDBank is the main target customer because it runs a tier-1 SCF portfolio at a national scale and was the first Vietnamese bank to join the Contour trade-finance blockchain network in 2020 (Emerio Banque 2024). The primary distribution channel is HDBank's tier-2/3 supplier pipeline; the secondary expansion vector is VNBA's Trade Finance Registry working group (VNBA 2026b).

The sales motion is land-and-expand: the HDBank contract is small and gated by a 90-day time-to-value window, with the consortium expanding once product adoption and champion engagement signals appear (Weber 2024). Performance is tracked through time-to-first-value, time-to-expand, CAC per bank, and net revenue retention.

3. PHASED TIMELINE

The pilot plan will expand over a 24-month period. Phase 1 (months 1-6) will focus on regulatory entry, applying for VIFC membership and a sandbox license under Decrees 323/2025 and 324/2025, endorsement from VNBA, and signing the HDBank MOU (VIFC Insight 2026; VTN and Partners 2025).

Phase 2 (months 7-12) deploys the five-component stack with HDBank as sole financing bank and onboards around 20 tier-2/3 suppliers. The combined Y1 funding gap (Phase 1 and 2) is about USD 192K (Appendix A2.5).

Phase 3 (months 13-18) onboards two to three additional banks and ships Layer 2 progressive disclosure, an AI Nudge Engine, ONCHAINID cross-bank claim reuse, and an ERC-4626 syndication vault. The Phase 3 gate is three or more active banks and HDBank advance volume above USD 2 million inside the 90-day time-to-value window (Weber 2024).

Phase 4 (months 19-24) adds a sealed-bid Marketplace, Pedersen-Bulletproof confidential amounts, DECO oracle privacy, and a tier-4 cascade extension. The Phase 4 gate is VIFC sandbox graduation plus SBV/VNBA endorsement letter.

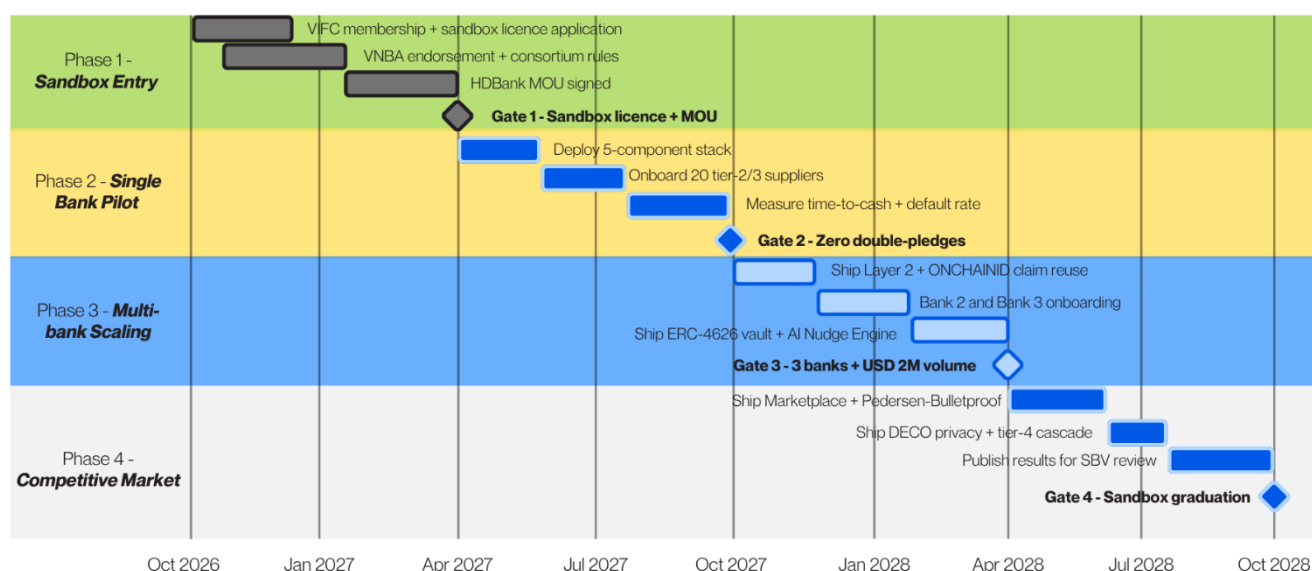


Figure 3: 24-month pilot Gantt

4. RISK MITIGATION

Phase	Risk	Mitigation
1	HDBank withdrawal at the lead-candidate stage	Signed MOU gates 1, locking HDBank commitment before Phase 2 spend. VNBA endorsement opens a backup pathway to alternative lead banks from the TFR pool (VNBA 2026b).
2	HDBank credit committee maintains conservative underwriting	The 90-day time-to-value contract gate forces measurable underwriting commitment (Weber 2024). HDBank's 2025 syndicated loan data-transparency conditions create top-down pressure to deploy into deep-tier MSME segments (VNBA 2025).
3	Bank-side adoption gap	HDBank's Contour 2020 precedent and the 90-day time-to-value gate reduce technology hesitation for incoming banks (Emerio Banque 2024; Weber 2024).
4	Marketplace cold start	Phase 3's multi-bank rhythm pre-populates the marketplace with three or more active banks. VNBA TFR's eight-bank density shows the threshold is achievable in the Vietnamese sector (VNBA 2026b).

Table 4: Phased risk mitigation

5. SCALABILITY AND SUSTAINABILITY

Adding a bank requires a validator node, consortium registration, and credit-scoring API integration. QBFT networks of four to eight validators tolerate up to two unresponsive nodes while keeping immediate finality, and throughput scales with validator count (Hyperledger Foundation 2026). The cascade-settlement mechanism extends to tier-3 and beyond without redesign, since each tier is merely a token transfer event (Dong et al. 2023).

Regarding sustainability, the subscription floor stabilizes revenue against transaction volatility. Consortium governance aligns rule changes with participant interest, preventing the profit-conflict that has eroded operator-led platforms. SBV audit access then provides forensic accountability against the reputational collapse pattern that ended TradeLens (Najati 2025).

VII. FEEDBACK ANALYSIS

1. INDUSTRY PANEL FEEDBACK

The industry panel provided feedback on the Assessment 2 stack over-engineered. The simplification was accepted, but the reasoning differed from the panels. The enforcement layer (Besu chain ERC-3643 token, Chainlink oracle,...) cannot be removed without losing fraud prevention. The syndication vault, ZK privacy, and marketplace can be moved to Phases 3 and 4. The twelve-component design dropped to a five-component Phase 2 stack. The evidence of cut track failure is not a generic call for less. we.trade blockchained the old flow without changing it and collapsed in 2022 (Tapprest 2023). Marco Polo signed thirty banks but recorded limited transaction volume before folding (Wragg 2023a). Ruff (2024) reads the pattern as trade needing protocols, not platforms.

2. INSTRUCTOR FEEDBACK

The instructor's comment highlighted the need for prototype-to-pain fit, the value proposition relative to existing models, and a detailed testing plan. In response, the report has improved as detailed in sections 5.1, 5.3, and 6.3, respectively.

Regarding prototype-to-pain fit, every component and its mechanisms of stack are already mapped to the corresponding pain, like the Chainlink functions to invoice authenticity, ERC-3643 to double-pledging, ONCHAINID for credit visibility, etc. This creates a stronger idea flow for the report

For the value proposition, the protocol is being compared to TReDS India and NAFIN Mexico. This has brought out the protocol's slot is neutral verification under consortium governance, which neither occupies.

And finally, there is a detailed testing plan, piloting for 24 months across four phases, with each phase gated on specific metrics: Phase 1 on VIFC conditional approval plus signed HDBank MOU, Phase 2 on the five-component stack live with 20 suppliers under HDBank, Phase 3 on three active banks and USD 2M advance volume within the 90-day time-to-value window, and Phase 4 on VIFC sandbox graduation plus SBV/VNBA endorsement.

VIII. SDG16 DISCUSSION

Regarding target 16.5, corruption reduction, the corresponding pattern in supply chain finance is double-financing fraud, where the same receivable is pledged multiple times (UN n.d.). The ERC-3643 standard is used to enforce token uniqueness; therefore, the receivable cannot exist on two ledgers, and the validator-bank quorum replaces single-issuer trust with multi-party consensus, creating auditable trails.

Also in line with UN target 16.6, effective and accountable institutions, SBV operates a supervisory node with read-only access to settlement events (UN n.d.). Auer (2019) describes this as “embedded supervision”, where compliance is machine-read directly from the ledger rather than reconstructed from intermediary reports. VIFC retains license-revocation authority across Phase 1, and consortium voting prevents the single-party drift.

On target 16.10, public access to information (UN 2015), receivable ownership is not made publicly visible, which would breach commercial privacy, but stays cross-institutionally auditable inside the consortium and supervisory perimeter. This is the realistic evidence-and-enforceability layer Nikolakis et al. (2018) identify in permissioned blockchain rather than full public disclosure.

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APPENDICES

Appendix A: Pilot Economics (Available through Excel file attached)

Appendix A1: Per-deal Economics: Mr. Thang Ngo VND 200 million invoice

A1.1 Trad SCF baseline (no protocol, supplier uses informal financing)

A1.1 Trad SCF baseline (no protocol, supplier uses informal financing)				
Line item	VND	USD	% of face	Source / formula
Invoice face value	200,000,000	\$8,000	100.00%	Mr. Thang Ngo reference deal
Tenor (days)	90			Typical Vietnam SCF deal
Informal financing cost (rate x tenor x face)	12,000,000	\$480	6.00%	Monthly informal rate x tenor in months x face
Net cash to supplier on day 1	188,000,000	\$7,520	94.00%	Face value minus informal cost
Effective annualised cost to supplier			24.0%	Monthly informal rate x 12 months

A1.2 Protocol-enabled path

Line item	VND	USD	% of face	Source / formula
Invoice face value	200,000,000	\$8,000	100.00%	Mr. Thang Ngo reference deal
Tenor (days)	90			
Bank discount rate to supplier (% p.a.)			12.0%	SBV system-average lending band plus tier-2/3
Bank gross interest income		\$237	2.96%	face x annual rate x tenor/365
Less: bank cost of fund	(2,958,904)	(\$118)	(1.48%)	face x annual fund cost x tenor/365
Less: bank risk provision	(1,000,000)	(\$40)	(0.50%)	Bank credit policy
Less: bank ops cost	(500,000)	(\$20)	(0.25%)	Salesforce and processing
Bank gross margin before protocol fee	1,458,904	\$58	0.73%	Sum of bank rows above
Protocol take rate (Y1 bps of face)	(240,000)	(\$10)	(0.12%)	Solomon (2015) 'rules and rails' Y1 pricing
Bank net margin after protocol fee	1,218,904	\$49	0.61%	Bank kept margin
Protocol fee as % of bank gross margin			16.5%	Protocol take rate divided by bank gross
Net cash to supplier on day 1 (protocol)	194,082,192	\$7,763	97.04%	Face value minus bank discount
Supplier saving vs informal baseline	6,082,192	\$243	3.04%	Protocol net cash minus informal net cash

A1.3 Distribution of value created on a single deal

A1.3 Distribution of value created on a single deal			
Stakeholder	Value captured (VND)	Value captured (USD)	Share of total economic surplus
Supplier (Mr. Thang Ngo)	6,082,192	\$243	80.65%
Bank net margin	1,218,904	\$49	16.16%
Protocol operator	240,000	\$10	3.18%
Risk provision and ops (bank reserves)	1,500,000	\$60	Reserve, not surplus
Total economic surplus released	7,541,096	\$302	100.00%

Appendix A2: Annual profit and loss in three scenario

A2.1 Volumn and participant assumptions

Parameter	Y1 Conservative	Y2 Base	Y3 Optimistic
Calendar year	2026	2027	2028
Stage	Sandbox pilot	Post-pilot expansion	Scale beyond sandbox
Participating banks	2	5	7
Anchor buyers	1	3	5
Active suppliers (midpoint)	8	40	125
Total deals per year	50	200	500
Average deal size (USD)	\$100,000	\$120,000	\$150,000
Total facilitated volume (USD)	\$5,000,000	\$24,000,000	\$75,000,000

A2.2 Revenue lines

Revenue line	Formula	Y1 Conservative	Y2 Base	Y3 Optimistic
Take rate revenue	Volume x take rate	\$6,000	\$33,600	\$112,500
Subscription revenue	Banks x subscription tier	\$30,000	\$125,000	\$245,000
Total revenue	Take + subscription	\$36,000	\$158,600	\$357,500
Revenue per facilitated USD	Total revenue divided by volume	0.72%	0.66%	0.48%

A2.3 Cost lines

Cost line	Y1 Conservative	Y2 Base	Y3 Optimistic
Smart contract audit	\$100,000	\$15,000	\$15,000
Besu node hosting	\$17,000	\$17,000	\$22,000
Chainlink Functions calls	\$1,000	\$5,000	\$12,000
Smart contract maintenance	\$20,000	\$20,000	\$22,000
Integration support engineer	\$40,000	\$40,000	\$45,000
Compliance and legal	\$20,000	\$20,000	\$22,000
Operations	\$30,000	\$30,000	\$35,000
Total cost	\$228,000	\$147,000	\$173,000

A2.4 Net result and break-even

Line	Y1 Conservative	Y2 Base	Y3 Optimistic
Total revenue	\$36,000	\$158,600	\$357,500
Total cost	(\$228,000)	(\$147,000)	(\$173,000)
Net	(\$192,000)	\$11,600	\$184,500
Cumulative net (running total)	(\$192,000)	(\$180,400)	\$4,100

A2.5 Funding gap and stakeholder ask

Year	Loss to cover	Cumulative funding required	Primary source
Y1	\$192,000	\$192,000	VIFC sandbox grant plus IFC advisory plus seed
Y2	\$11,600	\$203,600	Self-funding from revenue
Y3	\$184,500	\$388,100	Self-funding

Appendix A3: Sensitive analysis

A3.1 Take rate revenue matrix: deal volume x take rate

Deals per year	Volume (USD)	8 bps	12 bps	15 bps	20 bps
50	\$5,000,000	\$4,000	\$6,000	\$7,500	\$10,000
100	\$10,000,000	\$8,000	\$12,000	\$15,000	\$20,000
200	\$20,000,000	\$16,000	\$24,000	\$30,000	\$40,000
300	\$30,000,000	\$24,000	\$36,000	\$45,000	\$60,000
500	\$50,000,000	\$40,000	\$60,000	\$75,000	\$100,000
1,000	\$100,000,000	\$80,000	\$120,000	\$150,000	\$200,000

A3.2 Subscription revenue matrix: bank count × subscription tier

Participating banks	USD 10K tier	USD 15K tier	USD 25K tier	USD 35K tier
2	\$20,000	\$30,000	\$50,000	\$70,000
3	\$30,000	\$45,000	\$75,000	\$105,000
5	\$50,000	\$75,000	\$125,000	\$175,000
7	\$70,000	\$105,000	\$175,000	\$245,000
10	\$100,000	\$150,000	\$250,000	\$350,000

A3.3 Combined break-even surface

Deals per year (volume)	2 banks @ USD 15K	3 banks @ USD 20K	5 banks @ USD 25K	7 banks @ USD 30K
50 deals (USD 5,000,000)	\$36,000	\$66,000	\$131,000	\$216,000
100 deals (USD 10,000,000)	\$42,000	\$72,000	\$137,000	\$222,000
200 deals (USD 20,000,000)	\$54,000	\$84,000	\$149,000	\$234,000
500 deals (USD 50,000,000)	\$90,000	\$120,000	\$185,000	\$270,000
1,000 deals (USD 100,000,000)	\$150,000	\$180,000	\$245,000	\$330,000

A3.4 Single-variable shock tests

Scenario	Trigger	Revenue impact	Net Y2 result
Take rate compression	Regulator caps at 8 bps	Take rate USD 33,600 to USD 19,200	USD 158,600 to USD 144,200 (slight loss USD 2.8K vs cost USD 147K)
Bank attrition	1 of 5 banks drops out	Subscription USD 125K to USD 100K	USD 158,600 to USD 133,600 (slight loss USD 13K)
Audit cost overrun	Y1 audit USD 100K becomes USD 180K	Y1 loss USD 192K to USD 272K	Funding stack must expand USD 200K to USD 280K

Appendix B: Empathy and Define from Assignment 2

Appendix B1: Three customer personas

B1.1 Tier-2/3 supplier Persona: Mr. Thang Ngo

persona suppliers

Personal Details



Mr. Thang Ngo

(1) Owner-operator, 35-50 years old, 10-50 employees, revenue USD 300K-1.6M^{[1][2]}

(2) Sits 2-3 tiers below the anchor buyer in Bac Ninh or Dong Nai industrial zones^[3]

(3) Finance on Excel or paper, no ERP; only 10.5% of Vietnamese SMEs have ERP^[4]

Goals

- Convert confirmed POs into cash in 3-7 days, not 75-120^[5]
- Reach the formal bank rate of 8-9% instead of 24-36% informal^[6]
- Build a portable credit history that follows them across banks^[7]

Frustrations

- Structural 45-90 day cash gap every production cycle^{[8][9]}
- Collateral wall plus chain invisibility: bank demands real estate and cannot see the anchor 2-3 tiers up^{[8][10]}
- Informal lenders at 2-3% per month consume up to 40% of annual profit^{[8][11]}

Frustrations

- Accepts 75-120 day payment terms to preserve the chain position^[5]
- Pledges family home; borrows from moneylenders at 24-36% when not^{[8][12]}
- Extends more trade credit to own customers to stay competitive^[13]

Key Insights

(1) 40%+ of 5-year SMEs have never had a loan^[7]

- Verification infrastructure is the binding constraint, not creditworthiness

(2) 58% use accounting software, only 10.5% use ERP^[4]

- Protocol must integrate with MISA, Fast, and Excel, not displace them

(3) Suppliers will expose transaction data when trust, speed, and concrete gain are on offer^[10]

- Data-sharing willingness is conditional, not absolute

[1] (Bao Dai, Tu 2021), [2] (VITAS/Vietnam 2020), [3] (ADB/BAFT 2024), [4] (Nguyen et al. 2025), [5] (Lo 2026), [6] (Nhen Dan 2025), [7] (FinGroup 2025), [8] (Tap Chi Cong Thuong 2005), [9] (Entomew 2026), [10] (M/Economy 2025), [11] (Tan Phong/VCD 2026), [12] (UK Home Office 2025), [13] (Racine 2025)

B1.2 Bank lender Persona: Mr. Phuong

persona *banks*

Personal Details



Mr. Phuong

(1) ~55, Director of Corporate Banking, HDBank Executive Committee^{[1][2]}

(2) 28+ years banking across Standard Chartered, Citibank, BNP Paribas, Techcombank, TPBank, OCB^{[2][3]}

(3) Oversees 7,000+ businesses, 100 anchors, 4,202 tier-1 suppliers, VND 40,000B SCF portfolio^[1]

Goals

- Scale SCF from VND 40,000B to USD 3B, from 7,000 to 8,000 businesses^[3]
- Extend the three-party model from tier-1 down to tier-2 and tier-3^{[1][4]}
- Satisfy IFC, JICA, SECO data transparency and ESG conditions as funding terms^{[3][5]}

Frustrations

- Tier-1 depth cap: anchor ERPs end at tier-1, no digital link deeper^{[1][4]}
- Double-pledging and forged invoices undetectable at scale, no shared registry^[3]
- Circular 20/2024 creates a framework but not the tools; 80% of SMEs still manage finance manually^{[1][6]}

Frustrations

- Publicly frames SCF initiative as bank + anchor + SME on a shared data platform^[1]
- Embeds structurally with IFC, JICA, SECO, and SMBC as funders and advisors^{[3][5]}
- Speaks at VnEconomy Forum, FCI/IFC/VNBA International Factoring Seminar, Vietnam Investment Forum^{[2][3]}

Key Insights

(1) Publicly endorses the exact strategic model our protocol implements^[1]

- Lowering the conceptual selling burden

(2) HDBank's tier-1 book (35%+ annual growth, VND 40,000B) proves organizational capability^{[1][3]}

- The tier-2/3 extension is the missing technical layer

(3) Adoption depends on governance terms, tech fit, and vs-in-house-build competition

[1] (VnEconomy, 2025), [2] (Banker.vn, 2023), [3] (Entrepreneur, 2026), [4] (ADB/BAFT 2024), [5] (IFC-HDBank MoU 2022), [6] (VnEconomy, 2026)

B1.3 Anchor buyer Persona: Mr. Tuan Trinh

persona *anchor*

Personal Details



Mr. Tuan Trinh

- Senior PA Manager, FMCG export manufacturer, 45-60 years old^[1]
- Sources from multiple tier-2/3 suppliers across product categories; payments managed via corporate bank portal^{[1][6]}
- ERP covers tier-1 contracts only; tier-2/3 managed on paper and spreadsheets; only 10.5% of Vietnamese SMEs use ERP^[1]

Goals

- Zero delivery gaps to foreign MNC buyers: one missed shipment triggers penalty clauses (1-5% of contract value) and risks losing the relationship entirely^[1]
- Meet global HQ ESG requirements: supply chain traceability for labor conditions and financial health by 2026-27 is becoming a contract prerequisite^{[1][4]}
- Become preferred buyer: financially stable suppliers prioritize orders, accept tighter terms, and fulfill surge demands on short notice^{[1][6]}

Frustrations

- Supplier cash crises are invisible: suppliers blame material shortages rather than financial problems and the anchor discovers the real cause only after the deadline passes^{[1][6]}
- Forced to keep 2-3 backup suppliers per product category: a permanent operational cost of financial risk they cannot observe any other way^[1]
- Commercial confidentiality blocks data sharing: NDA clauses in every contract — invoice data exposes pricing strategy to competitors^[1]

Behaviors

- Extends payment to 60-90 days when own cash is tight, knowing it forces suppliers to informal debt at 24-36% — no alternative tool exists^[1]
- Offers early payment only as a bilateral price-reduction trade with tier-1 ("pay you early if you drop price by X%") — no mechanism reaches tier-2/3^{[1][9]}
- Delegates ESG compliance downstream via tier-1 pass-through clauses; zero verification capacity below that tier^{[1][4]}

Key Insights

- 2-3 backup supplier contracts kept idle permanently^[1]

» Supplier financial opacity is a direct operational cost to the anchor — not a social concern

- SME borrowing rate 15-18% vs anchor credit rate 8-10%^{[1][6]}

» DTSCF converts supplier interest savings into COGS negotiation leverage for the anchor

- CFO + CEO decide adoption; Procurement has no authority^[1]

» Benefit-first is the only pitch sequence that works — ROI before governance

[1] (Chia et al. 2024), [2] (Nguyen et al. 2025), [3] (Vietnam Briefing 2026), [4] (ILO 2025), [5] (Atradius 2025), [6] (ADB, 2023), [7] (Kiesing 2026), [8] (Amberg et al. 2024), [9] (Kovach et al. 2023), [10] (Giannetti et al. 2021)

Appendix B2: Empathy maps

B2.1 Tier-2/3 supplier Empathy map

supplier empathy

Think & Feel

- Anxious: the order is confirmed, but the cash to fill it is not^[1]
- Invisible to the banking system despite years of steady delivery: 40%+ of 5-year SMEs have never gotten a loan^[2]
- Trapped: no credit → no growth → no collateral (40.7% no capital accumulation)^{[3][4]}
- Formal credit is 'for other people'^{[5][12]}

See

- Confirmed POs sitting next to bank rejections^[5]
- Larger peers in the same zone run ERP and hold bank credit lines they cannot reach^[6]
- Moneylenders as a normal, visible fixture of the industrial zone^[7]

Hear

- 'No real estate, no chain verification, no loan'^{[1][8]}
- '60-120 day payment terms are the industry standard'^{[9][10][12]}
- 'Disbursement speed and bank trust are the binding constraints, not interest rates'^[11]

Say & Do

- 'I have Samsung's orders on my desk but the bank won't lend me money to fill them' (Minh/ tier-2)^{[13][2]}
- Accepts 75-120 day payment terms without negotiating, to preserve the chain position^{[9][12]}
- Pledges family home, then falls to moneylenders at 24-36% when the collateral wall is hit^{[3][7]}

[1] (E)nternew 2020; [2] (F)inGroup 2025; [3] (N)ien Den 2025; [4] (T)ien Phong/VCCI 2020; [5] (VCCI PCI 2022; [6] (N)guyen et al. 2025; [7] (B)K Home Office 2025; [8] (Vn)Economy 2025; [9] (L)u 2020; [10] (T)ap Chi Cong Thuong 2020; [11] (C)albitz 2020; [12] (A)ppendix A



B2.2 Bank lender Empathy map

bank empathy

Think & Feel

- Confident the SCF direction is correct: 'the bank's profit margin in this model is thinner than traditional lending, but the credit portfolio is safer, more efficient, and sustainable long-term'^[1]
- Frustrated that a technology gap caps the portfolio at tier-1: 'most anchors' ERP and procurement systems end at Tier 1'^[2]
- Worried one fraud event could undermine the entire SCF thesis: forged invoices and duplicate-pledged receivables are the top-cited risks^[3]

See

- 4,202 tier-1 suppliers, 100 anchors, VND 40,000B outstanding, 35%+ annual growth but stalling at tier-1 depth^{[1][3]}
- Peer banks remaining collateral-first: 80% excluded for insufficient collateral^[4]
- IFC, JICA, and SMBC partners who have already funded blockchain SCF globally: USD 215M syndicated loan to HDBank in 2025^{[5][3]}

Hear

- IFC/JICA: data transparency, ESG alignment, and digital infrastructure are preconditions for continued financing^[5]
- Risk team: double-pledging is undetectable without a shared registry^[3]
- Legal: Circular 20/2024 created the factoring framework but 'practical implementation still faces many obstacles'^{[6][8]}

Say & Do

- 'In the supply chain, the debt repayment cash flow of small suppliers actually comes from the large anchor enterprise. When the bank has this data, credit risk decreases significantly'^[1]
- Frames every SCF initiative as bank + anchor + SME on a shared data platform across FCI, IFC, VNBA, VnEconomy forums^{[10][8]}
- Accepts thinner SCF margins for portfolio quality, validated by actual book construction^{[11][7]}

[1] (Vn)Economy 2025; [2] (ADB/BAFT 2024; [3] (E)nternew 2020; [4] (VCCI PCI 2022; [5] (IFC+HDBank MoU 2023; [6] (Vn)Economy 2025; [7] (T)inh Nam 2025; [8] (B)ankers 2023



B2.3 Anchor buyer Empathy map

anchor empathy



Think & Feel

- Accountable for delivery SLAs but blind to tier-2/3 financial health: carries the risk without the information to manage it ^{[1][6]}
- WIN-WIN is the non-negotiable design principle: any cooperation that imposes cost on one side breaks; SCF tools that require the anchor to sacrifice working capital to help suppliers fail this test ^{[1][8]}
- Benefit before trust: will engage on governance and data sharing only after ROI is concretely proven ^{[1][7]}
- Wants to be the buyer suppliers choose to work with — cannot back that intention with any financial tool today ^{[1][8]}



Hear

- Suppliers (privately, outside formal meetings): "Can you pay us 30 days early? We're short on materials" ^{[6][5]}
- CFO: "Shortening payment terms is not an option — it would affect our working capital position" ^{[7][8]}
- Global HQ: "ESG traceability is a contractual prerequisite for EU market access by 2027" ^{[3][4]}



See

- Suppliers asking for early payment "between meetings" — the coded signal that informal debt is already the real problem ^{[6][5]}
- 2-3 idle backup supplier contracts per product category, maintained as a permanent hedge against invisible financial risk ^[9]
- Global HQ ESG reporting templates with tier-2/3 fields that cannot be filled — verified data ends at tier-1 ^{[3][4]}



Say & Do

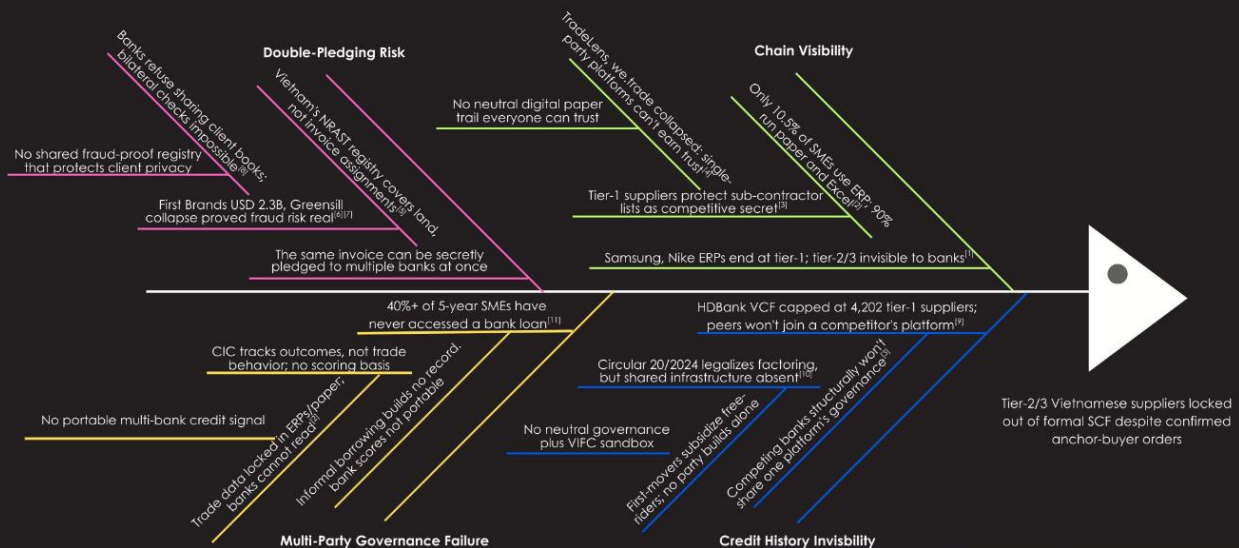
- "Suppliers experiencing financial distress do not report it — buyers learn of the problem only when the delivery is already late" ^{[6][5]}
- Negotiates early payment only as a price-reduction trade with tier-1 — no mechanism or budget for tier-2/3 ^{[8][10]}
- Signs supplier "mutual growth" partnership MOUs with no financing infrastructure below tier-1 to back them ^{[8][9]}

[1] (Chen et al. 2024), [2] (Vietnam Briefing 2026), [4] (ILO 2025), [5] (Alradus 2025), [6] (ADB 2022), [7] (Kosling 2026), [8] (Amberg et al. 2024), [9] (Kowach et al. 2023), [10] (Giannetti et al. 2024), [11] (TuringCorp 2024)



Appendix B3: Fishbone Analysis

fishbone analysis



[1] (ADB/BAFT 2024), [2] (Nguyen et al. 2025), [3] (Shibuya and Babich 2019), [4] (Bui et al. 2025), [5] (Ruslan and Vechy 2023), [6] (GTR 2025), [7] (Dodd 2024), [8] (Entrenew 2026), [9] (VnEconomy, 2025), [10] (VnEconomy 2026), [11] (FinGroup 2025)



Appendix B4: Problem Statement

problem statement



Problem Statement

Vietnamese tier-2/3+ suppliers to anchor companies struggle to convert **purchase orders/receivables** into working capital. Instead, they resort to informal loans at 24-36% interest, consuming up to 40%+ of their profits. While **capital and factoring regulations exist, banks can't reach these suppliers** due to issues such as invoice verification and a lack of credit history.



<https://www.scdprotocol.org/>



Appendix B5: How might we questions

how might we?

Problem

How Might We?

0 1

Make the Supply Chain Verifiable

- (1) How might we turn a confirmed PO into a token that the bank can verify?
- (2) How might we disclose only the chain linkage a bank needs, nothing more?
- (3) How might we keep anchor ERP data private while still proving the link?

0 2

Anchor Tier-3+ Supplier Financing

- (1) How might we prove an invoice is pledged only once, without revealing who holds it?
- (2) How might we let every bank verify instantly without a central operator?
- (3) How might we make the registry tamper-proof?

0 3

Enable Multi-Bank Cooperation

- (1) How might we make rival banks share a platform that no one controls?
- (2) How might we protect early participants under the VIFC regulatory sandbox?
- (3) How might we turn Circular 20/2024 into multi-bank factoring?

0 4

Score Deep-Tier Suppliers

- (1) How might we score a supplier from chain behavior, not loan history?
- (2) How might we make that score portable across every consortium bank?
- (3) How might we price risk fairly for tier-3 with no credit file?

<https://www.scdprotocol.org/>



Appendix B6: Customer Journey Map

